

IO List - RO Plant 1 & 2

Wilmar Industrial Complex PT MNA Serang, Indonesia

End Customer: Wilmar Industrial Complex PT MNA
Serang, Indonesia

End Customer Reference Number:

Document File Name: 3258585-PSS-PASH-DWG-00048
Document Status: For Final

Authorisations

	Name	Function	Signature
Developed By:	Zul Hasyidi Hassan	Project Engineer	ZHH 
Reviewed By:	Liow Kok Loong	Lead Engineer	LKL 
Released By:	Mohd. Nazim Ismail	Project Manager	MNI 

Distribution

Name	Company	Name	Company

Revision History

Rev	Status	Description	Date	Developed By	Reviewed By
A	For Review	Issued for Review	9-Nov-18	ZHH	LKL
0	For Final	Issued for Final	27-Sep-19	ZHH	LKL

© Emerson Automation Solution. All rights reserved. Unauthorized duplication, in whole or in part, is prohibited. Trademarks identified in this document are owned by one of the Emerson Process Management group of companies. Unless otherwise agreed to in writing by the parties, any information provided in this document is confidential or proprietary and may not be used or disclosed without the expressed written permission of Emerson Process Management.

Emerson Automation Solution Process System and Solutions

No. 1, Block A, Jalan SS13/5, SS13
Subang Jaya, 47500, Selangor
Malaysia
T +60 (3) 5624-2888
F +60 (3) 5638-1010
www.EmersonProcess.com

										RIO CABINET										SYSTEM CABINET						
										Marshalling						Relay		DELTA V				DELTA V				
Vendor's tags	Wilmar's Tags	Emerson's Tag	Module Name	Description	Type	I/O Type	Range Proposed (kg/H)	Signal type	Electrical Connection	Cabinet Name	TB Name	Term. (+)	Term. (-)	24V(+) Term.	24V(-) Term.	Term. (+)	Term. (-)	Name	Term. (+)	Term. (-)	Slot	Channel	Term	NA	Cabinet Name	Controller
LT06	LT1546	WP1-LT1546	WP1_LT1546	RO Storage Tank	Ultrasonic level transmitter	AI	0-4700mmH2O (0-100%)	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	59	60						S2	16	31	32	UT1-DCS-002	UTI-CTRL2		
LT07	LT1557	WP1-LT1557	WP1_LT1557	Demin Storage Tank	Ultrasonic level transmitter	AI	0-4700mmH2O (0-100%)	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	61	62						S3	1	1	2	UT1-DCS-002	UTI-CTRL2		
LT08	LT1538	WP1-LT1538	WP1_LT1538	UF CIP Tank	Ultrasonic level transmitter	AI	0-4700mmH2O (0-100%)	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	63	64						S3	2	3	4	UT1-DCS-002	UTI-CTRL2		
LT09	LT1549	WP1-LT1549	WP1_LT1549	RO CIP Tank	Ultrasonic level transmitter	AI	0-4700mmH2O (0-100%)	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	65	66						S3	3	5	6	UT1-DCS-002	UTI-CTRL2		
UT01	LT1551	WP1-LT1551	WP1_LT1551	Acid Bulk tank Level	Ultrasonic level transmitter	AI	0-4700mmH2O (0-100%)	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	67	68						S3	4	7	8	UT1-DCS-002	UTI-CTRL2		
UT02	LT1552	WP1-LT1552	WP1_LT1552	Caustic Bulk tank Level	Ultrasonic level transmitter	AI	0-4700mmH2O (0-100%)	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	69	70						S3	5	9	10	UT1-DCS-002	UTI-CTRL2		
UT03	LT1553	WP1-LT1553	WP1_LT1553	Acid metering tank Level control	Ultrasonic level transmitter	AI	0-4700mmH2O (0-100%)	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	71	72						S3	6	11	12	UT1-DCS-002	UTI-CTRL2		
UT04	LT1554	WP1-LT1554	WP1_LT1554	Caustic metering tank Level control	Ultrasonic level transmitter	AI	0-4700mmH2O (0-100%)	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	73	74						S3	7	13	14	UT1-DCS-002	UTI-CTRL2		
PID1A	PT1541A	WP1-PT1541A	WP1_PT1541A	RO 1 Array 1 Feed Pressure	Pressure Transmitter	AI	0-20 barg	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	75	76						S3	8	15	16	UT1-DCS-002	UTI-CTRL2		
PID2A	PT1542A	WP1-PT1542A	WP1_PT1542A	RO 1 Array 2 Feed Pressure	Pressure Transmitter	AI	0-20 barg	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	77	78						S3	9	17	18	UT1-DCS-002	UTI-CTRL2		
PID3A	PT1543A	WP1-PT1543A	WP1_PT1543A	RO 1 Concentrate Pressure	Pressure Transmitter	AI	0-20 barg	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	79	80						S3	10	19	20	UT1-DCS-002	UTI-CTRL2		
PID1B	PT1541B	WP1-PT1541B	WP1_PT1541B	RO 2 Array 1 Feed Pressure	Pressure Transmitter	AI	0-20 barg	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	81	82						S3	11	21	22	UT1-DCS-002	UTI-CTRL2		
PID2B	PT1542B	WP1-PT1542B	WP1_PT1542B	RO 2 Array 2 Feed Pressure	Pressure Transmitter	AI	0-20 barg	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	83	84						S3	12	23	24	UT1-DCS-002	UTI-CTRL2		
PID3B	PT1543B	WP1-PT1543B	WP1_PT1543B	RO 2 Concentrate Pressure	Pressure Transmitter	AI	0-20 barg	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	85	86						S3	13	25	26	UT1-DCS-002	UTI-CTRL2		
DP01	PD1521	WP1-PD1521	WP1_PD1521	Sand filter 1 & 2 - Differential pressure	Differential Pressure Transmitter	AI		4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	87	88						S3	14	27	28	UT1-DCS-002	UTI-CTRL2		
DP02	PD1522	WP1-PD1522	WP1_PD1522	Sand filter 3 & 4 - Differential pressure	Differential Pressure Transmitter	AI		4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	89	90						S3	15	29	30	UT1-DCS-002	UTI-CTRL2		
LT-95001	LT95001	WP1-LT95001	WP1_LT95001	Water Level Transmitter	Ultrasonic Level Transmitter	AI	0 - 17000 mm	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	91	92						S3	16	31	32	UT1-DCS-002	UTI-CTRL2		
LT-90191	LT90191	WP1-LT90191	WP1_LT90191	Water Level Transmitter	Ultrasonic Level Transmitter	AI	0 - 17000 mm	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	93	94						S3	17	33	34	UT1-DCS-002	UTI-CTRL2		
LT-1501A	LT1501A	WP1-LT1501A	WP1_LT1501A		Ultrasonic Level Transmitter	AI			2 wire	WP1-RIO-001	TB-AI	95	96						S4	2	3	4	UT1-DCS-002	UTI-CTRL2		
LT-1501B	LT1501B	WP1-LT1501B	WP1_LT1501B		Ultrasonic Level Transmitter	AI			2 wire	WP1-RIO-001	TB-AI	97	98						S4	3	5	6	UT1-DCS-002	UTI-CTRL2		
LT-102	LT102	WP1-LT102	WP1_LT102		Ultrasonic Level Transmitter	AI			2 wire	WP1-RIO-001	TB-AI	99	100						S4	4	7	8	UT1-DCS-002	UTI-CTRL2		
LT-103A	LT103A	WP1-LT103A	WP1_LT103A		Ultrasonic Level Transmitter	AI			2 wire	WP1-RIO-001	TB-AI	101	102						S4	5	9	10	UT1-DCS-002	UTI-CTRL2		
LT-103B	LT103B	WP1-LT103B	WP1_LT103B		Ultrasonic Level Transmitter	AI			2 wire	WP1-RIO-001	TB-AI	103	104						S4	6	11	12	UT1-DCS-002	UTI-CTRL2		
LT-90193	LT90193	WP1-LT90193	WP1_LT90193	Water Level Transmitter	Ultrasonic Level Transmitter	AI	0 - 17000 mm	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	105	106						S4	7	13	14	UT1-DCS-002	UTI-CTRL2		
LT-90195	LT90195	WP1-LT90195	WP1_LT90195	Water Level Transmitter	Ultrasonic Level Transmitter	AI	0 - 17000 mm	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	107	108						S4	8	15	16	UT1-DCS-002	UTI-CTRL2		
LT-0001	LT0001	WP1-LT0001	WP1_LT0001	Demin Water Level Transmitter	Ultrasonic Level Transmitter	AI	0 - 17000 mm	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	109	110						S4	9	17	18	UT1-DCS-002	UTI-CTRL2		
LT-0002	LT0002	WP1-LT0002	WP1_LT0002	Reject Water Level Transmitter	Ultrasonic Level Transmitter	AI	0 - 17000 mm	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	111	112						S4	10	19	20	UT1-DCS-002	UTI-CTRL2		
TTN1	TTN1	WP1-TTN1	WP1_TTN1	Temperature Transmitter for WTP MVD panel Room	Ultrasonic Level Transmitter	AI	0 - 100 DegC	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	113	114						S4	11	21	22	UT1-DCS-002	UTI-CTRL2		
TTN2	TTN2	WP1-TTN2	WP1_TTN2	Temperature Transmitter for WTP LVD panel Room	Ultrasonic Level Transmitter	AI	0 - 100 DegC	4-20 mA HART	2 wire	WP1-RIO-001	TB-AI	115	116						S4	12	23	24	UT1-DCS-002	UTI-CTRL2		
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-001	TB-AI	117	118						S4	13	25	26	UT1-DCS-002	UTI-CTRL2		
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-001	TB-AI	119	120						S4	14	27	28	UT1-DCS-002	UTI-CTRL2		
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-001	TB-AI	121	122						S4	15	29	30	UT1-DCS-002	UTI-CTRL2		
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-001	TB-AI	123	124						S4	16	31	32	UT1-DCS-002	UTI-CTRL2		
FE04	FT1524	WP1-FT1524	WP1_FT1524	Backwash water flowrate	+GF+ Signet Flow Transmitter c/w	AI	0-147 l/hr	4-20 mA HART	4 wire	WP1-RIO-001	TB-AI	125	126	17 +	17 -				S5	1	1	2	UT1-DCS-002	UTI-CTRL2		
FE05A	FT1535A	WP1-FT1535A	WP1_FT1535A	UF feed flowrate	+GF+ Signet Flow Transmitter c/w	AI	0-147 l/hr	4-20 mA HART	4 wire	WP1-RIO-001	TB-AI	127	128	18 +	18 -				S5	2	3	4	UT1-DCS-002	UTI-CTRL2		
FE06A	FT1536A	WP1-FT1536A	WP1_FT1536A	UF product flowrate	+GF+ Signet Flow Transmitter c/w	AI	0-147 l/hr	4-20 mA HART	4 wire	WP1-RIO-001	TB-AI	129	130	1												

										RIO CABINET										SYSTEM CABINET						
										DELTA V										DELTA V						
										DELTA V										DELTA V						
Vendor's tags	Wilmar's Tags	Emerson's Tag	Module Name	Description	Type	I/O Type	Range Proposed (kg/H)	Signal type	Electrical Connection	Cabinet Name	TB Name	Term. (+)	Term. (-)	24V(+) Term.	24V(-) Term.	Term. (+)	Term. (-)	Relay	Term. (+)	Term. (-)	Slot	Channel	Term	NA	Cabinet Name	Controller
4.XV-S05	V1525G-ZO	WP1-V1525G-ZO	WP1_V1525G	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-002	TB-DI	137	138								S3	5	5	NA	UTI-DCS-002	UTI-CTRL2
4.XV-S05	V1525G-ZC	WP1-V1525G-ZC	WP1_V1525G	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-002	TB-DI	139	140								S3	6	6	NA	UTI-DCS-002	UTI-CTRL2
4.XV-C01	V1521H-ZO	WP1-V1521H-ZO	WP1_V1521H	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-002	TB-DI	141	142								S3	7	7	NA	UTI-DCS-002	UTI-CTRL2
4.XV-C01	V1521H-ZC	WP1-V1521H-ZC	WP1_V1521H	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-002	TB-DI	143	144								S3	8	8	NA	UTI-DCS-002	UTI-CTRL2
4.XV-C02	V1522H-ZO	WP1-V1522H-ZO	WP1_V1522H	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-002	TB-DI	145	146								S3	9	9	NA	UTI-DCS-002	UTI-CTRL2
4.XV-C02	V1522H-ZC	WP1-V1522H-ZC	WP1_V1522H	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-002	TB-DI	147	148								S3	10	10	NA	UTI-DCS-002	UTI-CTRL2
4.XV-C03	V1523H-ZO	WP1-V1523H-ZO	WP1_V1523H	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-002	TB-DI	149	150								S3	11	11	NA	UTI-DCS-002	UTI-CTRL2
4.XV-C03	V1523H-ZC	WP1-V1523H-ZC	WP1_V1523H	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-002	TB-DI	151	152								S3	12	12	NA	UTI-DCS-002	UTI-CTRL2
4.XV-C04	V1524H-ZO	WP1-V1524H-ZO	WP1_V1524H	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-002	TB-DI	153	154								S3	13	13	NA	UTI-DCS-002	UTI-CTRL2
4.XV-C04	V1524H-ZC	WP1-V1524H-ZC	WP1_V1524H	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-002	TB-DI	155	156								S3	14	14	NA	UTI-DCS-002	UTI-CTRL2
4.XV-C05	V1525H-ZO	WP1-V1525H-ZO	WP1_V1525H	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-002	TB-DI	157	158								S3	15	15	NA	UTI-DCS-002	UTI-CTRL2
4.XV-C05	V1525H-ZC	WP1-V1525H-ZC	WP1_V1525H	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-002	TB-DI	159	160								S3	16	16	NA	UTI-DCS-002	UTI-CTRL2
FSI01	FSI1501	WP1-FSI1501	WP1_FSI1501	UFF backwash pump flow		DI		Saginomiya Flow switch		WP1-RIO-002	TB-DI	161	162								S3	17	17	NA	UTI-DCS-002	UTI-CTRL2
FSI02	FSI1502	WP1-FSI1502	WP1_FSI1502	RO permeate transfer pump flow		DI		Saginomiya Flow switch		WP1-RIO-002	TB-DI	163	164								S3	18	18	NA	UTI-DCS-002	UTI-CTRL2
FSI03	FSI1503	WP1-FSI1503	WP1_FSI1503	Mixed bed inlet flow		DI		Saginomiya Flow switch		WP1-RIO-002	TB-DI	165	166								S3	19	19	NA	UTI-DCS-002	UTI-CTRL2
LSL01	LSL1501	WP1-LSL1501	WP1_LSL1501	PAC chemical tank Level	Level Float switch	DI		Level Float switch	3 wire PNP	WP1-RIO-002	TB-DI	167	168	+		A1	A2(comm)	RDI-01	14	11(comm)	S3	20	20	NA	UTI-DCS-002	UTI-CTRL2
LSL02	LSL1502	WP1-LSL1502	WP1_LSL1502	NaOCl chemical tank Level	Level Float switch	DI		Level Float switch	3 wire PNP	WP1-RIO-002	TB-DI	169	170	+		A1	A2	RDI-02	14	11	S3	21	21	NA	UTI-DCS-002	UTI-CTRL2
LSL03	LSL1543	WP1-LSL1543	WP1_LSL1543	SMB5 chemical tank 01 Level	Level Float switch	DI		Level Float switch	3 wire PNP	WP1-RIO-002	TB-DI	171	172	+		A1	A2	RDI-03	14	11	S3	22	22	NA	UTI-DCS-002	UTI-CTRL2
LSL04	LSL1544	WP1-LSL1544	WP1_LSL1544	pH adjustment chemical tank 01 level	Level Float switch	DI		Level Float switch	3 wire PNP	WP1-RIO-002	TB-DI	173	174	+		A1	A2	RDI-04	14	11	S3	23	23	NA	UTI-DCS-002	UTI-CTRL2
LSL05	LSL1545	WP1-LSL1545	WP1_LSL1545	Antiscalant chemical tank 01 level	Level Float switch	DI		Level Float switch	3 wire PNP	WP1-RIO-002	TB-DI	175	176	+		A1	A2	RDI-05	14	11	S3	24	24	NA	UTI-DCS-002	UTI-CTRL2
LSL06	LSL1546	WP1-LSL1546	WP1_LSL1546	Biocide chemical tank 01 level	Level Float switch	DI		Level Float switch	3 wire PNP	WP1-RIO-002	TB-DI	177	178	+		A1	A2	RDI-06	14	11	S3	25	25	NA	UTI-DCS-002	UTI-CTRL2
LSL07	LSL1547	WP1-LSL1547	WP1_LSL1547	SMB5 chemical tank 02 Level	Level Float switch	DI		Level Float switch	3 wire PNP	WP1-RIO-002	TB-DI	179	180	+		A1	A2	RDI-07	14	11	S3	26	26	NA	UTI-DCS-002	UTI-CTRL2
LSL08	LSL1548	WP1-LSL1548	WP1_LSL1548	pH adjustment chemical tank 02 level	Level Float switch	DI		Level Float switch	3 wire PNP	WP1-RIO-002	TB-DI	181	182	+		A1	A2	RDI-08	14	11	S3	27	27	NA	UTI-DCS-002	UTI-CTRL2
LSL09	LSL1549	WP1-LSL1549	WP1_LSL1549	Antiscalant chemical tank 02 level	Level Float switch	DI		Level Float switch	3 wire PNP	WP1-RIO-002	TB-DI	183	184	+		A1	A2	RDI-09	14	11	S3	28	28	NA	UTI-DCS-002	UTI-CTRL2
LSL10	LSL1540	WP1-LSL1540	WP1_LSL1540	Biocide chemical tank 02 level	Level Float switch	DI		Level Float switch	3 wire PNP	WP1-RIO-002	TB-DI	185	186	+		A1	A2	RDI-10	14	11	S3	29	29	NA	UTI-DCS-002	UTI-CTRL2
LSL11	LSL1531	WP1-LSL1531	WP1_LSL1531	UF CIP tank level	Level Float switch	DI		Level Float switch	3 wire PNP	WP1-RIO-002	TB-DI	187	188	+		A1	A2	RDI-11	14	11	S3	30	30	NA	UTI-DCS-002	UTI-CTRL2
LSL12	LSL1542	WP1-LSL1542	WP1_LSL1542	RO CIP tank level	Level Float switch	DI		Level Float switch	3 wire PNP	WP1-RIO-002	TB-DI	189	190	+		A1	A2(comm)	RDI-12	14	11(comm)	S3	31	31	NA	UTI-DCS-002	UTI-CTRL2
FSI-103	FSI103	WP1-FSI103	WP1_FSI103	Water Distribution Flow Switch		DI				WP1-RIO-002	TB-DI	191	192								S3	32	32	NA	UTI-DCS-002	UTI-CTRL2
1.XV-S01	V1521A-SY	WP1-V1521A-SY	WP1_V1521A	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-002	TB-DO	1	NA			14	11	RDI-01	A1	A2(comm)	S6	1	1	NA	UTI-DCS-002	UTI-CTRL2
1.XV-S02	V1522A-SY	WP1-V1522A-SY	WP1_V1522A	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-002	TB-DO	2	NA			14	11	RDI-01	A1	A2	S6	2	2	NA	UTI-DCS-002	UTI-CTRL2
1.XV-S03	V1523A-SY	WP1-V1523A-SY	WP1_V1523A	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-002	TB-DO	3	NA			14	11	RDI-01	A1	A2	S6	3	3	NA	UTI-DCS-002	UTI-CTRL2
1.XV-S04	V1524A-SY	WP1-V1524A-SY	WP1_V1524A	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-002	TB-DO	4	NA			14	11	RDI-01	A1	A2	S6	4	4	NA	UTI-DCS-002	UTI-CTRL2
1.XV-S05	V1525A-SY	WP1-V1525A-SY	WP1_V1525A	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-002	TB-DO	5	NA			14	11	RDI-01	A1	A2	S6	5	5	NA	UTI-DCS-002	UTI-CTRL2
1.XV-C01	V1521B-SY	WP1-V1521B-SY	WP1_V1521B	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-002	TB-DO	6	NA			14	11	RDI-01	A1	A2	S6	6	6	NA	UTI-DCS-002	UTI-CTRL2
1.XV-C02	V1522B-SY	WP1-V1522B-SY	WP1_V1522B	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-002	TB-DO	7	NA			14	11	RDI-01	A1	A2	S6	7	7	NA	UTI-DCS-002	UTI-CTRL2
1.XV-C03	V1523B-SY	WP1-V1523B-SY	WP1_V1523B	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-002	TB-DO	8	NA			14	11	RDI-01	A1	A2	S6	8	8	NA	UTI-DCS-002	UTI-CTRL2
1.XV-C04	V1524B-SY	WP1-V1524B-SY	WP1_V1524B	Solenoid Valve c/w Limit Switch	Valve open command																					

										RIO CABINET										SYSTEM CABINET						
										Marshalling				Relay		DELTA V				DELTA V						
Vendor's tags	Wilmar's Tags	Emerson's Tag	Module Name	Description	Type	I/O Type	Range Proposed (kg/H)	Signal type	Electrical Connection	Cabinet Name	TB Name	Term. (+)	Term. (-)	24V(+) Term.	24V(-) Term.	Term. (+)	Term. (-)	Name	Term. (+)	Term. (-)	Slot	Channel	Term	NA	Cabinet Name	Controller
XV068	V15368-ZC	WP1-V15368-ZC	WP1_V15368	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	51	52								S1	26	26	NA	UTI-DCS-002	UTI-CTRL2
XV078	V15378-ZC	WP1-V15378-ZC	WP1_V15378	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	53	54								S1	27	27	NA	UTI-DCS-002	UTI-CTRL2
XV078	V15378-ZC	WP1-V15378-ZC	WP1_V15378	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	55	56								S1	28	28	NA	UTI-DCS-002	UTI-CTRL2
XV08A	V1548A-ZC	WP1-V1548A-ZC	WP1_V1548A	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	57	58								S1	29	29	NA	UTI-DCS-002	UTI-CTRL2
XV08A	V1548A-ZC	WP1-V1548A-ZC	WP1_V1548A	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	59	60								S1	30	30	NA	UTI-DCS-002	UTI-CTRL2
XV09A	V1549A-ZC	WP1-V1549A-ZC	WP1_V1549A	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	61	62								S1	31	31	NA	UTI-DCS-002	UTI-CTRL2
XV09A	V1549A-ZC	WP1-V1549A-ZC	WP1_V1549A	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	63	64								S1	32	32	NA	UTI-DCS-002	UTI-CTRL2
XV10A	V1540A-ZC	WP1-V1540A-ZC	WP1_V1540A	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	65	66								S2	1	1	NA	UTI-DCS-002	UTI-CTRL2
XV10A	V1540A-ZC	WP1-V1540A-ZC	WP1_V1540A	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	67	68								S2	2	2	NA	UTI-DCS-002	UTI-CTRL2
XV08B	V1548B-ZC	WP1-V1548B-ZC	WP1_V1548B	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	69	70								S2	3	3	NA	UTI-DCS-002	UTI-CTRL2
XV08B	V1548B-ZC	WP1-V1548B-ZC	WP1_V1548B	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	71	72								S2	4	4	NA	UTI-DCS-002	UTI-CTRL2
XV09B	V1549B-ZC	WP1-V1549B-ZC	WP1_V1549B	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	73	74								S2	5	5	NA	UTI-DCS-002	UTI-CTRL2
XV09B	V1549B-ZC	WP1-V1549B-ZC	WP1_V1549B	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	75	76								S2	6	6	NA	UTI-DCS-002	UTI-CTRL2
XV10B	V1540B-ZC	WP1-V1540B-ZC	WP1_V1540B	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	77	78								S2	7	7	NA	UTI-DCS-002	UTI-CTRL2
XV10B	V1540B-ZC	WP1-V1540B-ZC	WP1_V1540B	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	79	80								S2	8	8	NA	UTI-DCS-002	UTI-CTRL2
XV12	V1552A-ZC	WP1-V1552A-ZC	WP1_V1552A	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	81	82								S2	9	9	NA	UTI-DCS-002	UTI-CTRL2
XV12	V1552A-ZC	WP1-V1552A-ZC	WP1_V1552A	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	83	84								S2	10	10	NA	UTI-DCS-002	UTI-CTRL2
XV13	V1553A-ZC	WP1-V1553A-ZC	WP1_V1553A	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	85	86								S2	11	11	NA	UTI-DCS-002	UTI-CTRL2
XV13	V1553A-ZC	WP1-V1553A-ZC	WP1_V1553A	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	87	88								S2	12	12	NA	UTI-DCS-002	UTI-CTRL2
XV14	V1554A-ZC	WP1-V1554A-ZC	WP1_V1554A	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	89	90								S2	13	13	NA	UTI-DCS-002	UTI-CTRL2
XV14	V1554A-ZC	WP1-V1554A-ZC	WP1_V1554A	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	91	92								S2	14	14	NA	UTI-DCS-002	UTI-CTRL2
XV15	V1555A-ZC	WP1-V1555A-ZC	WP1_V1555A	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	93	94								S2	15	15	NA	UTI-DCS-002	UTI-CTRL2
XV15	V1555A-ZC	WP1-V1555A-ZC	WP1_V1555A	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	95	96								S2	16	16	NA	UTI-DCS-002	UTI-CTRL2
XV16	V1556-ZC	WP1-V1556-ZC	WP1_V1556	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	97	98								S2	17	17	NA	UTI-DCS-002	UTI-CTRL2
XV16	V1556-ZC	WP1-V1556-ZC	WP1_V1556	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	99	100								S2	18	18	NA	UTI-DCS-002	UTI-CTRL2
XV17	V1557-ZC	WP1-V1557-ZC	WP1_V1557	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	101	102								S2	19	19	NA	UTI-DCS-002	UTI-CTRL2
XV17	V1557-ZC	WP1-V1557-ZC	WP1_V1557	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	103	104								S2	20	20	NA	UTI-DCS-002	UTI-CTRL2
XV18	V1558-ZC	WP1-V1558-ZC	WP1_V1558	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	105	106								S2	21	21	NA	UTI-DCS-002	UTI-CTRL2
XV18	V1558-ZC	WP1-V1558-ZC	WP1_V1558	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	107	108								S2	22	22	NA	UTI-DCS-002	UTI-CTRL2
XV19	V1559-ZC	WP1-V1559-ZC	WP1_V1559	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	109	110								S2	23	23	NA	UTI-DCS-002	UTI-CTRL2
XV19	V1559-ZC	WP1-V1559-ZC	WP1_V1559	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	111	112								S2	24	24	NA	UTI-DCS-002	UTI-CTRL2
XV20	V1550-ZC	WP1-V1550-ZC	WP1_V1550	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	113	114								S2	25	25	NA	UTI-DCS-002	UTI-CTRL2
XV20	V1550-ZC	WP1-V1550-ZC	WP1_V1550	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	115	116								S2	26	26	NA	UTI-DCS-002	UTI-CTRL2
XV21	V1551-ZC	WP1-V1551-ZC	WP1_V1551	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	117	118								S2	27	27	NA	UTI-DCS-002	UTI-CTRL2
XV21	V1551-ZC	WP1-V1551-ZC	WP1_V1551	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	119	120								S2	28	28	NA	UTI-DCS-002	UTI-CTRL2
XV22	V1552B-ZC	WP1-V1552B-ZC	WP1_V1552B	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	121	122								S2	29	29	NA	UTI-DCS-002	UTI-CTRL2
XV22	V1552B-ZC	WP1-V1552B-ZC	WP1_V1552B	Solenoid Valve c/w Limit Switch	Valve close indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	123	124								S2	30	30	NA	UTI-DCS-002	UTI-CTRL2
XV23	V1553B-ZC	WP1-V1553B-ZC	WP1_V1553B	Solenoid Valve c/w Limit Switch	Valve open indication	DI		DI Dry Contact	2 wire	WP1-RIO-003	TB-DI	125	126								S2	31				

										RIO CABINET												SYSTEM CABINET				
										Marshalling				Relay				DELTA V				DELTA V		Cabinet Name	Controller	
Vendor's tags	Wilmar's Tags	Emerson's Tag	Module Name	Description	Type	I/O Type	Range Proposed (kg/H)	Signal type	Electrical Connection	Cabinet Name	TB Name	Term. (+)	Term. (-)	24V(+) Term.	24V(-) Term.	Term. (+)	Term. (-)	Name	Term. (+)	Term. (-)	Slot	Channel	Term	NA		
XV08A	V1548A-SY	WP1-V1548A-SY	WP1_V1548A	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-003	TB-DO	29	30			14	11	RDO1-13	A1	A2	S6	15	15	NA	UTI-DCS-002	UTI-CTRL2
XV09A	V1549A-SY	WP1-V1549A-SY	WP1_V1549A	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-003	TB-DO	31	32			14	11	RDO1-14	A1	A2	S6	16	16	NA	UTI-DCS-002	UTI-CTRL2
XV10A	V1540A-SY	WP1-V1540A-SY	WP1_V1540A	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-003	TB-DO	33	34			14	11	RDO1-14	A1	A2	S6	17	17	NA	UTI-DCS-002	UTI-CTRL2
XV08B	V1548B-SY	WP1-V1548B-SY	WP1_V1548B	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-003	TB-DO	35	36			14	11	RDO1-14	A1	A2	S6	18	18	NA	UTI-DCS-002	UTI-CTRL2
XV09B	V1549B-SY	WP1-V1549B-SY	WP1_V1549B	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-003	TB-DO	37	38			14	11	RDO1-14	A1	A2	S6	19	19	NA	UTI-DCS-002	UTI-CTRL2
XV10B	V1540B-SY	WP1-V1540B-SY	WP1_V1540B	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-003	TB-DO	39	40			14	11	RDO1-24	A1	A2	S6	20	20	NA	UTI-DCS-002	UTI-CTRL2
XV12	V1552A-SY	WP1-V1552A-SY	WP1_V1552A	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-003	TB-DO	41	42			14	11	RDO1-2	A1	A2	S6	21	21	NA	UTI-DCS-002	UTI-CTRL2
XV13	V1553A-SY	WP1-V1553A-SY	WP1_V1553A	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-003	TB-DO	43	44			14	11	RDO1-22	A1	A2	S6	22	22	NA	UTI-DCS-002	UTI-CTRL2
XV14	V1554A-SY	WP1-V1554A-SY	WP1_V1554A	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-003	TB-DO	45	46			14	11	RDO1-2	A1	A2	S6	23	23	NA	UTI-DCS-002	UTI-CTRL2
XV15	V1555A-SY	WP1-V1555A-SY	WP1_V1555A	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-003	TB-DO	47	48			14	11	RDO1-24	A1	A2	S6	24	24	NA	UTI-DCS-002	UTI-CTRL2
XV16	V1556-SY	WP1-V1556-SY	WP1_V1556	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-003	TB-DO	49	50			14	11	RDO1-24	A1	A2	S6	25	25	NA	UTI-DCS-002	UTI-CTRL2
XV17	V1557-SY	WP1-V1557-SY	WP1_V1557	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-003	TB-DO	51	52			14	11	RDO1-24	A1	A2	S6	26	26	NA	UTI-DCS-002	UTI-CTRL2
XV18	V1558-SY	WP1-V1558-SY	WP1_V1558	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-003	TB-DO	53	54			14	11	RDO1-24	A1	A2	S6	27	27	NA	UTI-DCS-002	UTI-CTRL2
XV19	V1559-SY	WP1-V1559-SY	WP1_V1559	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-003	TB-DO	55	56			14	11	RDO1-24	A1	A2	S6	28	28	NA	UTI-DCS-002	UTI-CTRL2
XV20	V1550-SY	WP1-V1550-SY	WP1_V1550	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-003	TB-DO	57	58			14	11	RDO1-24	A1	A2	S6	29	29	NA	UTI-DCS-002	UTI-CTRL2
XV21	V1551-SY	WP1-V1551-SY	WP1_V1551	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-003	TB-DO	59	60			14	11	RDO1-34	A1	A2	S6	30	30	NA	UTI-DCS-002	UTI-CTRL2
XV22	V1552B-SY	WP1-V1552B-SY	WP1_V1552B	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-003	TB-DO	61	62			14	11	RDO1-34	A1	A2	S6	31	31	NA	UTI-DCS-002	UTI-CTRL2
XV23	V1553B-SY	WP1-V1553B-SY	WP1_V1553B	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-003	TB-DO	63	64			14	11	RDO1-34	A1	A2	S6	32	32	NA	UTI-DCS-002	UTI-CTRL2
XV24	V1554B-SY	WP1-V1554B-SY	WP1_V1554B	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-003	TB-DO	65	66			14	11	RDO1-34	A1	A2	S7	1	1	NA	UTI-DCS-002	UTI-CTRL2
XV25	V1555B-SY	WP1-V1555B-SY	WP1_V1555B	Solenoid Valve c/w Limit Switch	Valve open command	DO		24VDC Wet Supply	2 wire	WP1-RIO-003	TB-DO	67	68			14	11	RDO1-34	A1	A2	S7	2	2	NA	UTI-DCS-002	UTI-CTRL2
SV-101A	V101A-SY	WP1-V101A-SY	WP1_V101A		Actuator Valve	DO				WP1-RIO-003	TB-DO	69	70			14	11	RDO1-34	A1	A2	S7	3	3	NA	UTI-DCS-002	UTI-CTRL2
SV-101B	V101B-SY	WP1-V101B-SY	WP1_V101B		Actuator Valve	DO				WP1-RIO-003	TB-DO	71	72			14	11	RDO1-34	A1	A2	S7	4	4	NA	UTI-DCS-002	UTI-CTRL2
SV-102	V102-SY	WP1-V102-SY	WP1_V102		Actuator Valve	DO				WP1-RIO-003	TB-DO	73	74			14	11	RDO1-34	A1	A2	S7	5	5	NA	UTI-DCS-002	UTI-CTRL2
SV-103A	V103A-SY	WP1-V103A-SY	WP1_V103A		Actuator Valve	DO				WP1-RIO-003	TB-DO	75	76			14	11	RDO1-34	A1	A2	S7	6	6	NA	UTI-DCS-002	UTI-CTRL2
SV-103B	V103B-SY	WP1-V103B-SY	WP1_V103B		Actuator Valve	DO				WP1-RIO-003	TB-DO	77	78			14	11	RDO1-34	A1	A2	S7	7	7	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	79	80			14	11	RDO1-44	A1	A2	S7	8	8	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	81	82			14	11	RDO1-44	A1	A2	S7	9	9	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	83	84			14	11	RDO1-44	A1	A2	S7	10	10	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	85	86			14	11	RDO1-44	A1	A2	S7	11	11	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	87	88			14	11	RDO1-44	A1	A2	S7	12	12	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	89	90			14	11	RDO1-44	A1	A2	S7	13	13	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	91	92			14	11	RDO1-44	A1	A2	S7	14	14	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	93	94			14	11	RDO1-44	A1	A2	S7	15	15	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	95	96			14	11	RDO1-44	A1	A2	S7	16	16	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	97	98			14	11	RDO1-44	A1	A2	S7	17	17	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	99	100			14	11	RDO1-54	A1	A2	S7	18	18	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	101	102			14	11	RDO1-54	A1	A2	S7	19	19	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	103	104			14	11	RDO1-54	A1	A2	S7	20	20	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	105	106			14	11	RDO1-54	A1	A2	S7	21	21	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	107	108			14	11	RDO1-54	A1	A2	S7	22	22	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	109	110			14	11	RDO1-54	A1	A2	S7	23	23	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	111	112			14	11	RDO1-54	A1	A2	S7	24	24	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	113	114			14	11	RDO1-54	A1	A2	S7	25	25	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	115	116			14	11	RDO1-64	A1	A2	S7	26	26	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	117	118			14	11	RDO1-54	A1	A2	S7	27	27	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	119	120			14	11	RDO1-64	A1	A2	S7	28	28	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	121	122			14	11	RDO1-64	A1	A2	S7	29	29	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	123	124			14	11	RDO1-64	A1	A2	S7	30	30	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	125	126			14	11	RDO1-64	A1	A2	S7	31	31	NA	UTI-DCS-002	UTI-CTRL2
SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	SPARE	WP1-RIO-003	TB-DO	127	128			14	11	RDO1-64	A1	A2(comm)	S7	32	32	NA	UTI-DCS-002	UTI-CTRL2